

Open Addressing

1. Concept

Definition

Open Addressing is a collision resolution technique in which all elements are stored directly inside the hash table itself.

Unlike chaining, no linked lists are used.

When a collision occurs, we search for another empty slot using a probing sequence.

2. General Probing Formula

$$h(k, i) = \text{Probe Function}$$

Where:

- k = key
- i = probe number ($i = 0, 1, 2, \dots$)
- m = table size

First attempt:

$$h(k, 0)$$

If occupied $\rightarrow$ try:

$$h(k, 1), h(k, 2), \dots$$

until an empty slot is found.

3. Example

Let:

$$m = 7$$

$$h(k) = k \mod 7$$

Insert keys:

10, 17, 24

$$h(10) = 3$$

$$h(17) = 3$$

$$h(24) = 3$$

All collide at index 3.

Open addressing finds new empty slots.

Visual Representation

0	1	2	3	4	5	6
			10	17	24	

Elements are placed in nearby empty slots.

4. Types of Open Addressing

- Linear Probing
- Quadratic Probing
- Double Hashing

Each method defines a different probe function.

5. Load Factor Condition

$$\alpha = \frac{n}{m}$$

Where:

- n = number of elements
- m = table size

Important Condition

For Open Addressing:

$$\alpha < 1$$

Table must have at least one empty slot.

If $\alpha \rightarrow 1$, performance degrades rapidly.

6. Advantages and Disadvantages

Advantages

- No extra memory for linked lists
- Better cache performance
- Simple array implementation

Disadvantages

- Clustering problem
- Performance sensitive to load factor
- Deletion is more complicated

Open Addressing = Compact Memory but Load Sensitive